



ELECT MAINT Department
SIMHADRI

Note No :1



Ref:SMTTP/ELECT MAINT/2023-24/LMI-EMD/373812

Date:

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Subject:LMI Revision -LMI/OGN/OPS/ELEC/19-Dissolved gas analysis of Transformers & Reactors

LMI Reference No:LMI/OGN/OPS/ELEC/19

LMIs are reviewed periodically for further improvement and incorporation of recommendation by COS.No changes envisaged by COS . LMI has been checked and reviewed for latest updates and checklist as per latest OGN .

Put up approval of competent authority please.

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NTPC LIMITED

SIMHADRI SUPER THERMAL POWER PLANT

LOCAL MANAGEMENT INSTRUCTION

LMI/OGN/OPS/ELEC/19

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Reviewed on 07.08.2023 and No revision in line with CCOS

DISSOLVED GAS ANALYSIS OF TRANSFORMERS

&

REACTORS

APPROVED FOR

IMPLEMENTATION _ _ _ _ _

CGM (SIMHADRI)

DATE _ _ _ _ _

NTPC LTD

SIMHADRI THERMAL POWER PLANT

OPERATION & MAINTENANCE DEPARTMENT

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DISSOLVED GAS ANALYSIS OF TRANSFORMERS & REACTORS

1.0 INTRODUCTION

Dissolved gas analysis (DGA) is a diagnostic technique in which by monitoring the trend of various fault gases in transformers and reactors oil, interpretation of incipient faults in these equipments can be made. For this the oil sampling for DGA has to be done on regular basis. There are many national and international standards and practices for interpretation for dissolved gas results based on which the guidelines have been prepared and described in subsequent paragraphs.

The guidelines given in this document are for (a) oil sampling frequency for carrying out dissolved gas analysis of transformers/reactors and (b) the method for interpretation of the fault gases found dissolved in oil.

2.0 SUPERSEDED DOCUMENT

LMI/OGN/OPS/ELEC/19, Issue-1, Rev-2, December 2019

3.0 RECORDS OF REVISION

NIL

4.0 REFERENCE DOCUMENT

- 1) COS-ISO-00-OGN/OPS/ELEC/019
- 2) IEEE Std C57-104-2008
- 3) IEC – 60599: 2015

5.0 SCOPE :

Guidelines are given in this document regarding oil sampling frequency for carrying out DGA. They are applicable for following Transformers at Simhadri:-

- i. GT : 1R, 1Y, 1B, 2R, 2Y, 2B, 3R,3Y,3B,4R,4Y,4B
- ii. Spare GT Stage #1 & 2
- iii. 400/11 kV S/B Transformer Stage #1&2
- iv. Unit Transformer 1A, 1B, 2A, 2B, 3 &4
- v. Unit Aux. Transformer 1A, 1B, 2A, 2B,3A, 3B, 4A & 4B
- vi. Reactor
- vii. 11/ 33 kV MST – A,B,&C
- viii. 33/ 11 kV Sea Water Transformers.
- ix. 33/ 0.415 kV Ash Water Re-Circulation Transformers.
- x. 33/0.415 kV Sweet Water Transformers
- xi. LT outdoor Service Transformer

6.0 GUIDELINES

6.01 Guidelines regarding oil sampling frequency for DGA of transformers and reactors:

The following schedules are recommended for DGA of the transformers / reactors.

(a) Transformers/Reactors (33 KV and above) & Unit Transformers:-

- | | |
|--------------------------|---|
| 1 st Sample - | Just prior to charging |
| 2 nd Sample – | within 24hrs to 48hrs of charging |
| 3 rd Sample- | 15 days after charging or as recommended. |
| 4 th Sample - | After 1 month of charging or as recommended |
| 5 th Sample - | After 3 months of charging or as recommended and continues 3 monthly for Generator Transformers and 6 monthly for other transformers and reactor or as recommended. |

Note:

Oil samples of the transformers and reactors to be sent to NETRA with above schedule for DGA. Oil sample of these transformers and reactors are to be tested with one month frequency at site or by out sourcing if DGA kit is not available at site. Deviated DGA results are to be forwarded to CC-OS for review and further advice.

(b) For Transformers upto 11 kV:-

1st Sample - 24hr after charging

2nd Sample – Annually or as recommended and continue annually for subsequent sample.

Oil sample of these transformers are to be tested at site or by out sourcing if DGA kit is not available at site. Deviated DGA results are to be forwarded to CC-OS for review and further advice.

6.02 Guidelines for study & interpretation of DGA results:

On day to day basis, all stations are to study the results and If fault gases in any transformer are found to be exceeding the limits or the trend of fault gases is rising significantly, then the case may immediately be referred to Corporate Operation Services for further analysis and guidance.

6.03 Guideline regarding on line DGA monitoring of transformers:

- a) Where ever on line DGA analyzer is installed on the transformers, the results are to be down loaded and viewed by stations regularly on fortnightly basis. On observation of deviation or significant rise in any fault gas, oil sample is to be tested immediately at site and repeat sample may be sent to NETRA for confirmation.
- b) The absolute values of fault gases from online DGA analyzer and that of NETRA may not match. In such cases only the trend of fault gases may be accounted for monitoring purpose.
- c) If the absolute values of the online DGA analyzer and that of NETRA & Site lab are not matching and the trend of fault gases is also inconsistent with wide variations, the matter may be taken up with on line DGA analyzer supplier / manufacturer to calibrate / rectify the same.

6.04 Information and guideline for DGA results in SAP:

- a) The results of DGA done by NETRA are to be uploaded in SAP. The results for individual transformers and exceptions can be seen by transaction (T) code ZPMDGA_REP / ZPNDGA_EX_REP.
- b) If any EHV transformer and critical transformers are not appearing in SAP for DGA results, the functional location and codes for the same may be generated immediately and sent to ERP for incorporation.

6.05 Guideline regarding DGA facility at site:

Dissolve Gas Analysis is a very important and vital technique to assess the condition of transformers in service. The technique helps to find out the initiation of fault in a transformer and corrective measures can be taken to avoid failure of the transformer. It also helps in planning the maintenance need for transformers during overhauling / shutdown of the Unit. Sometimes it may be very urgent to do DGA of a transformer on hourly basis to take decision regarding further serviceability of the transformer in operation. Therefore following may be adhere to:

- 1) All stations need to have laboratory model DGA instrument at their Chemistry lab for Dissolve Gas Analysis of transformers. The DGA instrument should be readily available at chemistry lab of the station for scheduled DGA of transformers and for any unscheduled requirements.
- 2) Gas based stations may have portable DGA instrument for Dissolve Gas Analysis of transformers as the quantity of transformers to be monitored is quite low in comparison to coal based stations.
- 3) DGA results of all transformer done at site or by out sourcing need to be uploaded on individual station website under chemistry section, Dissolved as analysis domain.

6.06 Guidelines for interpretation of fault gases:**a) Permissible dissolved gas level:**

For the interpretation of the results of DGA, two factors, namely, the rate of gas evolution and the concentration of individual fault gases are very important. Following table indicates the permissible limit of various fault gases in transformer. The values are for reference only. Individual fault gases below this level generally indicate normal operation of the transformer as per IEEE Std C57-104-2008. Rising trend in individual fault gases need to be observed closely to understand the fault and the parts involved.

Table-1 (Reference: IEEE Std C57-104-2008)

S.No	Name of the Gas	Chemical Symbol	<u>Permissible Levels</u> <u>(ppm V/V)</u>
1.	Hydrogen	H ₂	100
2.	Methane	CH ₄	120
3.	Acetylene	C ₂ H ₂	1
4.	Ethylene	C ₂ H ₄	50
5.	Ethane	C ₂ H ₆	65
6.	Carbon Monoxide	CO	350
7.	Carbon Di-oxide	CO ₂	2500

b) Evaluation of Transformer Condition Using Individual and TDCG Concentrations:**Table 2 (Reference: IEEE Std C57-104-2008)**

Status	Dissolved Key Gas Concentration Limits (ppm)							
	H ₂	CH ₄	C ₂ H ₂	C ₂ H ₄	C ₂ H ₆	CO	CO ₂	TDCG**
Condition-1	100	120	1	50	65	350	2500	720
Condition-2	101-700	121-400	2-9	51-100	66-100	351-570	2500-4000	721-1920
Condition-3	701-1800	401-1000	10-35	101-200	101-150	571-1400	4001-10000	1921-4630
Condition-4	>1800	>1000	>35	>200	>150	>1400	>10000	>4630

** The TDCG value does not include CO₂, which is not a combustible gas.

Condition 1 TDCG below this level indicates the transformer is operating satisfactorily. Any individual combustible gas exceeding specified levels should prompt additional investigation.

Condition 2 TDCG within this range indicates greater than normal combustible gas level. Any individual combustible gas exceeding specified levels should prompt additional investigation. Action should be taken to establish a trend. Fault(s) may be present.

Condition 3 TDCG within this range indicates a high level of decomposition. Any individual combustible gas exceeding specified levels should prompt additional investigation. Immediate action should be taken to establish a trend. Fault(s) are probably present.

Condition 4 TDCG within this range indicates excessive decomposition. Continued operation could result in failure of the transformer.

c) Types of faults (Reference: IEC – 60599: 2015)

Internal inspection of hundreds of faulty transformer has led to the following broad classes of visually detectable faults:

- Partial discharges (PD) of the cold plasma (corona) type, resulting in possible X-wax deposition on paper insulation, or of the sparking type, inducing pinhole, carbonized perforations (punctures) in paper, which, however, may not be easy to find;
- Discharges of low energy (D1), in oil or/and paper, evidenced by larger carbonized perforations through paper (punctures), carbonization of the paper surface (tracking) or carbon particles in oil (as in tap changer diverter operation);
- Discharges of high energy (D2), in oil or/and paper, with power follow-through, evidenced by extensive destruction and carbonization of paper, metal fusion at the discharge extremities, extensive carbonization in oil and, in some cases, tripping of the equipment, confirming the large current follow-through;
- Thermal faults, in oil or/and paper, below 300 °C if the paper has turned brownish (T1), and above 300 °C if it has carbonized (T2);
- Thermal faults of temperatures above 700 °C (T3) if there is strong evidence of carbonization of the oil, metal coloration (800 °C) or metal fusion (>1 000 °C).

Table 3 – Abbreviations (Reference: IEC – 60599: 2015)

PD	Partial discharges
D1	Discharges of low energy
D2	Discharges of high energy
T1	Thermal fault, $t < 300\text{ °C}$
T2	Thermal fault, $300\text{ °C} < t < 700\text{ °C}$
T3	Thermal fault, $t > 700\text{ °C}$

d) Basic gas ratios: (Reference: IEC – 60599: 2015)

Each of the six broad classes of faults leads to a characteristic pattern of hydrocarbon gas composition, which can be translated into a DGA interpretation table based on the use of three basic gas ratios:

**Table 4 – DGA Interpretation Table (Reference: IEC – 60599: 2015)**

CASE	FAULT TYPE	C_2H_2/C_2H_4	CH_4/H_2	C_2H_4/C_2H_6
PD	Partial discharges	NS	<0,1	<0,2
D1	Discharges of low energy	>1	0,1 – 0,5	>1
D2	Discharges of high energy	0,6 – 2,5	0,1 – 1	>2
T1	Thermal fault $t < 300\text{ }^{\circ}\text{C}$	NS	>1 but NS1)	<1
T2	Thermal fault $300\text{ }^{\circ}\text{C} < t < 700\text{ }^{\circ}\text{C}$	<0,1	>1	1 – 4
T3	Thermal fault $t > 700\text{ }^{\circ}\text{C}$	<0,22)	>1	>4
NOTE 1 – In some countries, the ratio C_2H_2/C_2H_6 is used, rather than the ratio CH_4/H_2 . Also in some countries, slightly different ratio limits are used.				
NOTE 2 – The above ratios are significant and should be calculated only if at least one of the gases is at a concentration and a rate of gas increase above typical values				
NOTE 4 – Gas decomposition patterns similar to partial discharges have been reported as a result of stray gassing				
NS = Non-significant whatever the value.				
An increasing value of the amount of C_2H_2 may indicate that the hot spot temperature is higher than $1000\text{ }^{\circ}\text{C}$.				

NOTE – Combinations of gas ratios which fall outside the range limits of table 4 and do not correspond to a characteristic fault of this table may be considered a mixture of faults, or new faults which combine with a high background gas level.

e) Gas Ratio Interpretation:

1) CO₂/CO ratio

The formation of CO₂ and CO from oil-impregnated paper insulation increases rapidly with temperature. High values of CO (e.g., 1 000 ppm) and CO₂/CO ratios less than 3 are generally considered as an indication of probable paper involvement in a fault, with possible carbonization, in the presence of other fault gases. However, in some recent transformers of the closed-type or open (free breathing) transformers operating at constant load (i.e., with low breathing), CO can accumulate in the oil, leading to ratio CO₂/CO < 3, without any irregularities or faults if no other gases such as H₂ or hydrocarbons are formed. High values of CO₂ (>10 000 ppm) and high CO₂/CO ratios (>10) can indicate mild (<160 °C) overheating of paper or oil oxidation, especially in open transformers. CO₂ can accumulate more rapidly than CO in open transformers operating at changing loads because of their different solubilities in oil. This, and the long term degradation with time of paper at low temperatures (<160 °C), can lead to higher CO₂/CO ratios in aged equipment. In some cases, localized faults in paper do not produce significant amounts of CO and CO₂ and cannot be detected with these gases (the same for furanic compounds). Involvement of faults in paper therefore shall not be based only on CO and CO₂, but shall be confirmed by the formation of other gases or other types of oil analysis.

In order to get reliable CO₂/CO ratios in the equipment, CO₂ and CO values should be corrected (incremented) first for possible CO₂ absorption from atmospheric air, and for the CO₂ and CO background values resulting from the ageing of cellulose insulation, overheating of wooden blocks and the long term oxidation of oil (which will be strongly influenced by the availability of oxygen caused by specific equipment construction details and its way of operation).

Air-breathing equipment, for example, saturated with approximately 10 % of dissolved air, may contain up to 300 µl/l of CO₂ coming from the air. In sealed equipment, air is normally excluded but may enter through leaks, and CO₂ concentration will be in proportion of air present. When excessive paper degradation is suspected (CO₂/CO < 3), it is advisable to ask for a furanic compounds analysis or a measurement of the degree of polymerization of paper samples, when this is possible.

2) O₂/N₂ ratio

Dissolved O₂ and N₂ may be found in oil, as a result of contact with atmospheric air in the conservator of air-breathing equipment, or through leaks in sealed equipment. At equilibrium, taking into account the relative solubilities of O₂ and N₂, the O₂/N₂ ratio in oil reflects air composition and is close to 0.5.

In service, this ratio may decrease as a result of oil oxidation and/or paper ageing, if O₂ is consumed more rapidly than it is replaced by diffusion. Factors such as the load and preservation system used may also affect the ratio, but ratios less than 0,3 are generally considered to indicate excessive consumption of oxygen.

3) C₂H₂/H₂ ratio

In power transformers, on load tap changer (OLTC) operations produce gases corresponding to discharges of low energy (D1). If some oil or gas communication is possible between the OLTC compartment and the main tank, or between the respective conservators, these gases may contaminate the oil in the main tank and lead to wrong diagnoses. The pattern of gas decomposition in the OLTC, however, is quite specific and different from that of regular D1s in the main tank.

C₂H₂/H₂ ratios higher than 2 to 3 in the main tank are thus considered as an indication of OLTC contamination. This can be confirmed by comparing DGA results in the main tank, in the OLTC and in the conservators. The values of the gas ratio and of the acetylene concentration depend on the number of OLTC operations and on the way the contamination has occurred (through the oil or the gas).

f) Application to free gases in gas relays:

During a fault, the production rate of gases of all types is closely linked to the rate of energy liberation. Thus, the low rate of energy liberation in partial discharges, or in a low-temperature hot spot, will cause gases to evolve slowly and there is every probability that all the gas produced will dissolve in the oil. The higher rate of energy liberation of a high-temperature core fault, for example, can cause an evolution of gas rapid enough to result in gas bubbles. These will usually partially dissolve in the oil (and exchange with gases already dissolved) but some gas may well reach the gas collecting relay or gas cushion; this gas may approach equilibrium with the gases dissolved in the oil.

In principle, the analysis of free gases from a gas collecting relay or from a gas cushion may be evaluated in the same way as the analysis of gases dissolved in the oil. However, where the surge element has operated and gas has accumulated in substantial quantities, there is a possibility of having a serious fault, and analyses of the gases should be undertaken to identify the fault. Buchholz alarms due to air accumulation are

also possible following a combination of warm days and sudden temperature drops at night. It is therefore important to collect the gas at the relay as soon as possible without burning it, and sample the oil in the relay and in the main tank.

Where gas has accumulated slowly, assessment of the gases dissolved in the oil is more informative than that of the free gases; this gas-in-oil analysis is also essential in order to determine the total rate of evolution of gases and thus check whether the fault is growing which is the most important matter to investigate. When analysis of free gases is undertaken, it is necessary to convert the concentrations of the various gases in Free State into equivalent concentrations in the dissolved state, using table 5, before applying the gas ratio method of table -4.

Applying the principles set out above, comparison of the actual concentrations in the oil with the equivalent concentrations in the free gas may give valuable information on how far gas bubbles may have risen through the oil and, hence, on the rate of gas evolution.

The calculation of dissolved gas concentrations equivalent to free gas concentrations is made by applying the Ostwald coefficient for each gas separately. For a particular gas, the Ostwald coefficient k is defined as follows:

$$\text{Ostwald co-efficient } K = \frac{\text{Concentration of gas in liquid space}}{\text{Concentration of gas in gas space}}$$

The Ostwald coefficient for various gases in the mineral oil at 20 C and 50C are given in table 5 below. **Reference IEC-60599)**

Gas	K at 20 C	K at 50 C
N ₂	0.09	0.09
O ₂	0.17	0.17
H ₂	0.05	0.05
CO	0.12	0.12
CO ₂	1.08	1.00
CH ₄	0.43	0.40
C ₂ H ₆	2.40	1.80
C ₂ H ₄	1.70	1.40
C ₂ H ₂	1.20	0.90

Table 5

The Ostwald coefficient is independent of the actual partial pressures of the gas concerned. The gas and liquid phases are assumed to be at the same temperature; this is rarely the case but the error introduced by any difference will not invalidate the conclusions reached.

g) Evaluation of possible fault type by the Doernenburg ratio method

Table 6—Ratios for key gases—Doernenburg								
Suggested fault diagnosis	Ratio 1 (R1)		Ratio 2 (R2)		Ratio 3 (R3)		Ratio 4 (R4)	
	CH₄/H₂		C₂H₂/C₂H₄		C₂H₂/CH₄		C₂H₆/C₂H₂	
	Oil	Gas space	Oil	Gas space	Oil	Gas space	Oil	Gas space
1. Thermal decomposition	>1.0	>0.1	<0.75	<1.0	<0.3	<0.1	>0.4	>0.2
2. Partial discharge (low-intensity PD)	<0.1	<0.01	Not significant		<0.3	<0.1	>0.4	>0.2
3. Arcing (high-intensity PD)	>0.1 to <1.0	>0.01 to <0.1	>0.75	>1.0	>0.3	>0.1	<0.4	<0.2

h) Evaluation of possible fault type by the Rogers ratio method

Table 7—Rogers ratios for key gases				
Case	R2 C₂H₂/C₂H₄	R1 CH₄/H₂	R5 C₂H₄/C₂H₆	Suggested fault diagnosis
0	<0.1	>0.1 to <1.0	<1.0	Unit normal
1	<0.1	<0.1	<1.0	Low-energy density arcing—PDa
2	0.1 to 3.0	0.1 to 1.0	>3.0	Arcing—High-energy discharge
3	<0.1	>0.1 to <1.0	1.0 to 3.0	Low temperature thermal
4	<0.1	>1.0	1.0 to 3.0	Thermal <700 °C
5	<0.1	>1.0	>3.0	Thermal >700 °C

i) Interpretation of results using Duval Triangle

Calculate percentage value of acetylene, ethylene and methane by following method:-

$$x = \text{C}_2\text{H}_2 \text{ in ppm}$$

$$y = \text{C}_2\text{H}_4 \text{ in ppm}$$

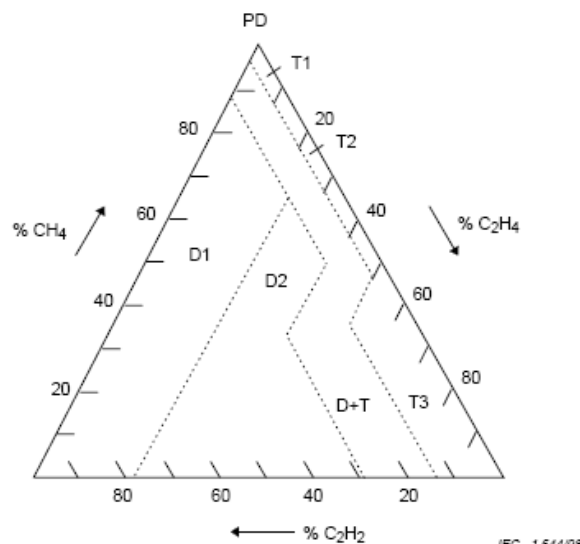
$$z = \text{CH}_4 \text{ in ppm}$$

$$\% \text{C}_2\text{H}_2 = \frac{100 x}{x+y+z} \%$$

$$\% \text{C}_2\text{H}_4 = \frac{100 y}{x+y+z} \%$$

$$\% \text{CH}_4 = \frac{100 z}{x+y+z} \%$$

After calculating the percentage value of the three fault gases, find out the block in the triangle below, where these gases are present. Table 8 below gives the type of fault expected inside the equipment based on the coordinates in the triangle:



DUVAL TRIANGLE

Table-8 (Reference:IEC-60599)

PD	98 % CH ₄			
D1	23 % C ₂ H ₄	13 % C ₂ H ₂		
D2	23 % C ₂ H ₄	13 % C ₂ H ₂	38 % C ₂ H ₄	29 % C ₂ H ₂
T1	4 % C ₂ H ₂	10 % C ₂ H ₄		
T2	4 % C ₂ H ₂	10 % C ₂ H ₄	50 % C ₂ H ₄	
T3	15 % C ₂ H ₂	50 % C ₂ H ₄		

j) Interpretation based on rate of rise of individual dissolved gases in oil **Table 9**

No.	Main gas Components	Rate of rise	Diagnosis	Action
1.	CH ₄ , C ₂ H ₄ , C ₂ H ₆	Less than 10% per month	Unsatisfactory condition Local heating in oil.	More frequent analysis is required to monitor any change of gas patterns and rate of rise
	If each of the above gas is > 150 PPM	More than 10% per month	Serious condition Local heating developing in oil	In addition to above, internal inspection is advisable at a convenient time for de energisation.
2.	C ₂ H ₂ , H ₂	Less than 10% per month	Serious situation Intermittent arc discharge in oil.	More frequent analysis and internal-inspection at an early stage
		More than 10% per month	Critical situation Continuous arc discharge in oil	Internal inspection is required as soon as possible
3	C ₂ H ₂ , H ₂ , CO	Less than 10% per month	Serious situation Intermittent arc discharge between coils	Electrical tests and internal inspection are required as soon as possible
		More than per 10% month	Very critical situation. Continuous discharge between coils	Quick deenergisation for electrical tests and internal inspection.

(Maintenance Guide by the analysis of dissolved gas in oil for transformers and

Reactors in service By MELCO-AKO works SNK-30655)

The electrical test results are very important for systematic diagnosis of a faulty transformer.

The advisable electrical tests are shown in Table 9A

S.No.	Test item	Local	Arc Discharge	Partial Discharge
1.	Insulation resistance	√	√	√
2.	Oil test (see note 1)	√	√	√
3.	Ratio Test	√	√	
4.	Exciting current (note 2)	√	√	
5.	Winding resistance		√	√
6.	Insulation power factor		√	√
7	Over-voltage withstand test			√
8	Partial discharge test			√
9	Temperature distribution on tank wall	√		
10	Vibration of tank wall	√		
11	Operation test of protective relays	√	√	√
12	Operation test of On-Load tap changer	√	√	
13	Operation test of oil pump motors	√	√	√
14	Analysis of gas accumulated in Buchholz	√	√	√

Maintenance guide by the analysis of dissolved gases in oil for transformers reactors in service by MELCO-AKO works)

Note: 1. Oil test means dielectric strength, acid value resistivity and moisture content

Note: 2. Exciting current may be measured at low voltage.

Note: 3. Tests marked with '√' may be performed for different faults as per Table 9.

k) Interpretation data of transformer fault gas by James E.Morgan

In the Table no. 1 there is no indication of either the age or the volume of the transformers to which the above values apply, even though it is recognized that

both of these parameters can influence fault gas levels. In view of the above, a guide for the interpretation data of transformer fault gases by James E.Morgan of National Power, shown in Table no. 10 gives most practical and safe fault gas levels in oil immersed transformers.

Table 10

(Extract from Paper “A guide to the Interpretation data of transformer fault gas” data by James E.Morgan)

Sl.No.	Name of the gas	Chemical Symbol	Permissible dissolved gas concentration (ppm V/V)	
	Hydrogen	H ₂	Less than	20n+50
	Methane	CH ₄		20n+50
	Ethane	C ₂ H ₆		20N+50
	Ethylene	C ₂ H ₄		20n+50
	Accetylene	C ₂ H ₂		5n+10
	Carbon Monoxide	CO		25n+500
	Carbon Dioxide	CO ₂		100n+1500
	Total fault gases			110n+710

n = Number of years in service

6.07 Guideline on transformer oil aging parameter testing:

Following Oil parameters are to be tested on six monthly basis for all EHV and Critical Unit Related transformers.

- Oil Tan Delta (limit: 0.2 / 0.5 max at 90°C for >170kV / <170kV CL)
- Oil Resistivity (0.1X10¹² Ohm-cm at 90°C)
- Oil Acidity (0.25 mg KOH/gm)
- Oil BDV (> 60kV)
- Moisture content in oil (10ppm for EHV class & 15 ppm for others)

Note: IFT (Inter Facial Tension) & Flash point of transformer oil are to be measured if one of above parameters at a), b) &c) violate limits.

7.0 Specific Instructions To Be Followed:

- a) Samples to be collected as per frequency given in this document ahead.
- b) Sample frequency may vary as per the advice of R&D Lab/ NOIDA or site Chemistry Lab.

8.0 RESPONSIBILITY CENTRES FOR IMPLEMENTATION:

INSTRUCTION	RESPONSIBILITY
Oil Sampling and sending to labs.	Area Maint. Engineer (EMD)
Electrical tests required	Area Maint. Engineer (EMD)
Testing of Samples and interpretation	R&D Noida/ Chemistry lab at site

9.0 EXCEPTION REPORTING & DISPENSATION OF DEVIATION:

All exceptions would be brought to the notice of HOD-Electrical Maintenance and can be dispensed with his approval.

10.0 RECORD KEEPING:

All records and reports received from R&D Lab & Site Chemistry Lab would be kept in LMI Files of Electrical maintenance area groups.

11.0 REVIEW

Head of Project, Simhadri will be responsible for review of this document on two yearly basis.